Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 6

Title: Sales Data Analysis and Sorting and Filtering Data in Power BI

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Sales Data Analysis and Sorting and Filtering Data in Power BI

1. Problem Statement

The objective of this experiment is to analyze sales data using Microsoft Power BI by applying sorting and filtering techniques and creating interactive visualizations. The analysis focuses on identifying the highest-revenue transactions, extracting high-value transactions, identifying underperforming products, and evaluating overall sales performance using KPI cards, charts, and slicers to support business decision-making.

2. Methodology / Steps Followed

Step 1: Import the Dataset

- Opened Power BI Desktop.
- Imported the sales analysis dataset from an Excel file.
- Checked the columns and data types required for analysis.

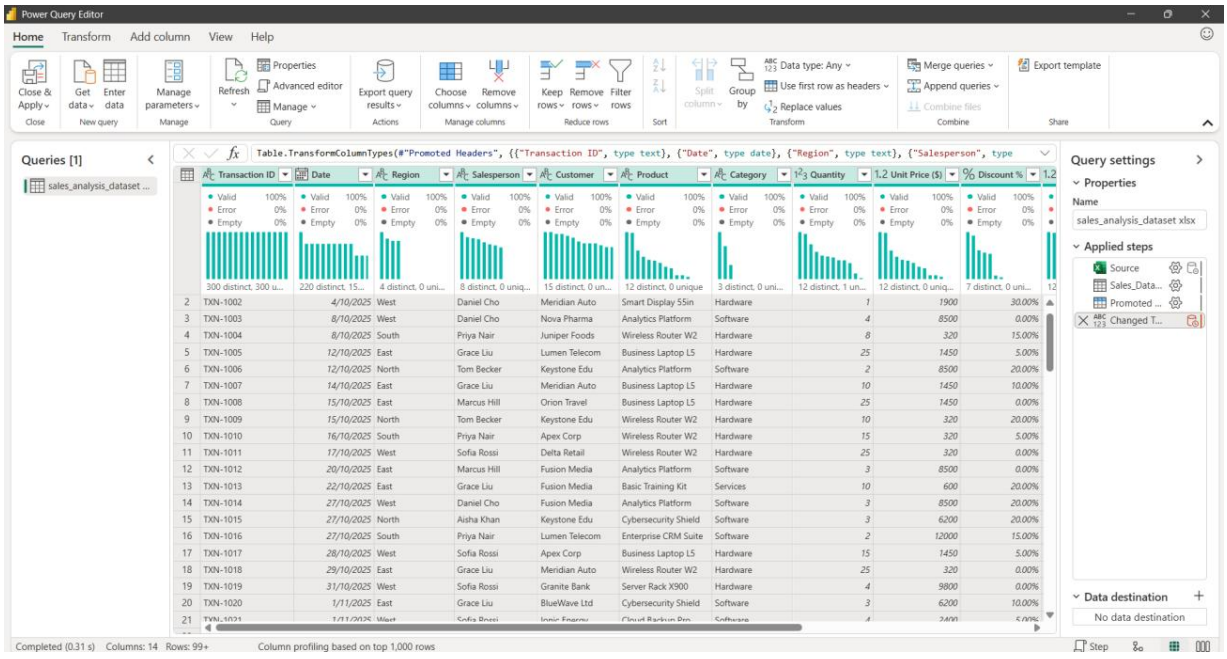


Figure 1: Imported the Sales Data into Power BI

Step 2: Prepare the Data

- Reviewed the available sales data.
- Checked and corrected data types where necessary.
- Identified relevant fields such as Transaction ID, Region, Product, Category, Revenue, Profit, and Margin.

- Ensured that the data was ready for analysis and visualization.

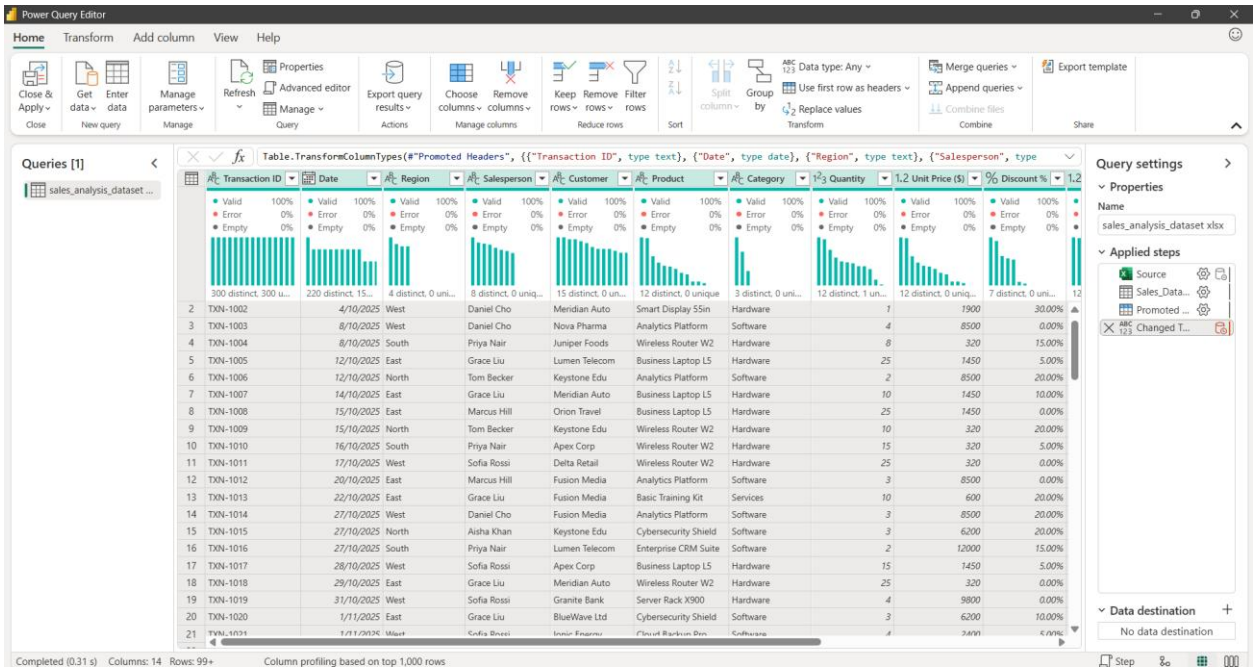


Figure 2: Checking for the Datatype of a Column

Step 3: Create the Top Transactions Table

- Created a table visual containing transaction details and revenue.
- Applied a Top N filter to identify high-value transactions.
- Sorted the revenue column in descending order.
- Identified the transactions that generated the highest revenue.

Step 4: Filter High-Value Transactions

- Created a separate table visual.
- Applied a revenue filter of greater than or equal to $32,400.
- Identified the transactions that satisfied this condition.
- Displayed the filtered results in a table for analysis.

Step 5: Identify Underperforming Products

- Created a table to analyze product performance.
- Applied the relevant filter to display underperforming products.
- Sorted revenue in ascending order to identify products with the lowest revenue first.
- Examined revenue and margin to identify products requiring further business review.

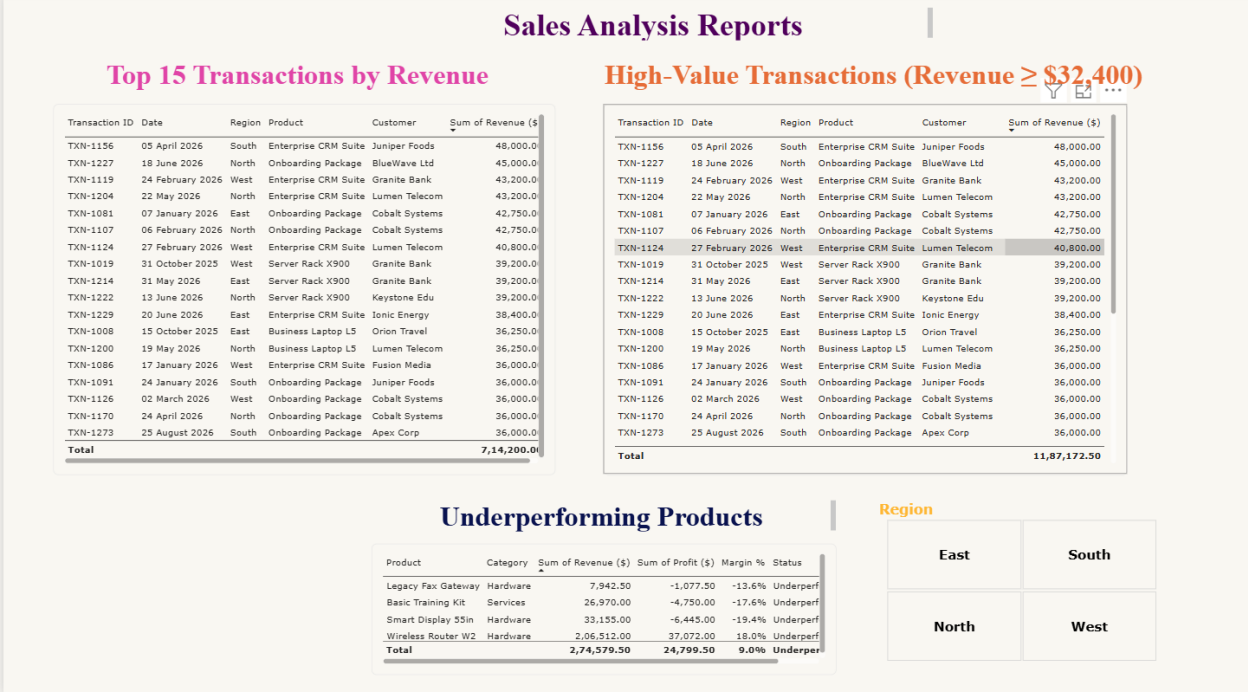


Figure 3: Sales Analysis Dashboard , Top 15 Transactions by Revenue High-Value Transactions (Revenue ≥ $32,400) , Underperforming Products

Step 6: Create KPI Cards for Sales Analysis

Created KPI cards to summarize important sales performance measures.

- **Total Profit:** Displays the total profit earned from sales.
- **Total Revenue:** Displays the overall revenue generated.
- **Profit Margin:** Displays the profit margin as a percentage.
- **Average Revenue per Product:** Displays the average revenue associated with products.
- **Average Profit per Product:** Displays the average profit associated with products.
- **Total Transactions:** Displays the total number of transactions.

These KPI cards provide a quick overview of business performance and make it easier to evaluate sales and profitability.

Step 7: Create a Clustered Bar Chart

- Inserted a clustered bar chart in Power BI.
- Used Product as the category or axis field.
- Used Total Revenue as the value field.
- Compared revenue generated by different products.
- Used the chart to identify products contributing more or less revenue.

Step 8: Create a Pie Chart for Category-Wise Revenue

- Inserted a pie chart in Power BI.
- Used Category as the category or legend field.
- Used Total Revenue as the value field.
- Displayed the contribution of each category to total revenue.
- Compared the relative revenue contribution of different categories.

Step 9: Add a Region Slicer

- Added a slicer visual to the dashboard.
- Selected Region as the slicer field.
- Enabled users to select a specific region.
- Used the slicer to filter the dashboard visuals and analyze sales performance for the selected region.

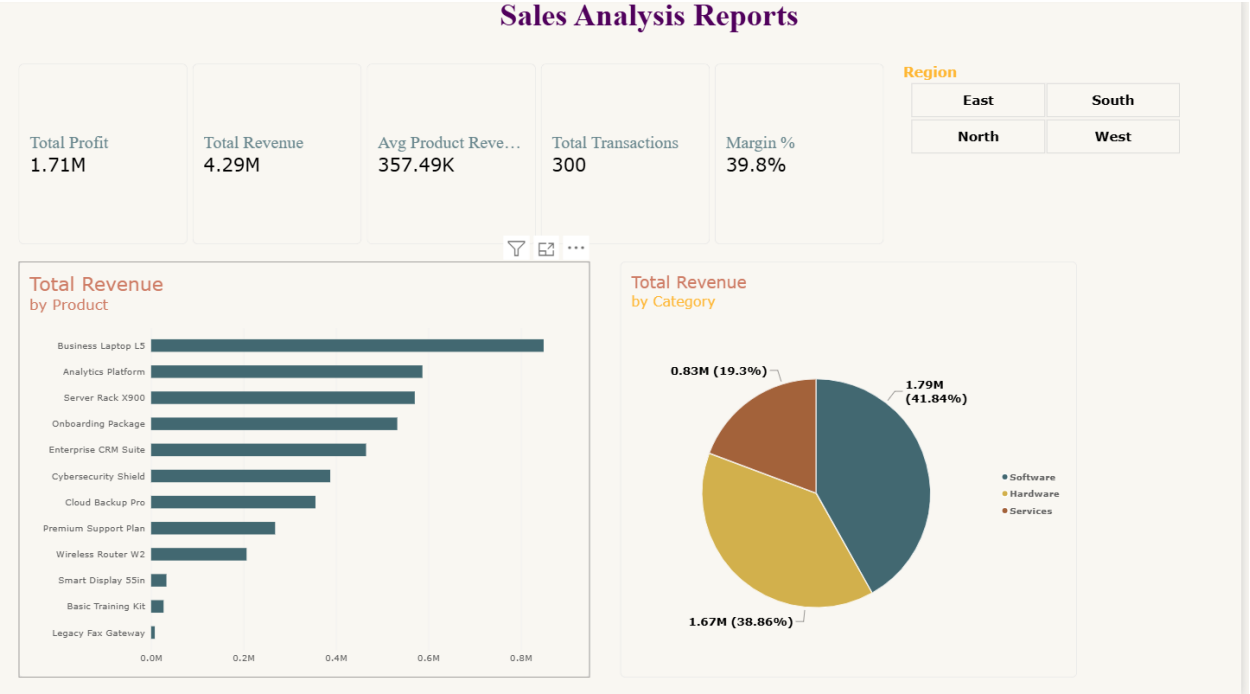


Figure 4: ***Sales Analysis Dashboard – Sorting and Filtering includes KPI card, Charts***

3. Output

The experiment produced two Power BI dashboard pages.

Dashboard 1: Sorting and Filtering Analysis

The dashboard contains the following tables:

1. **Top 15 Transactions by Revenue:** Displays the transactions with the highest revenue.
2. **High-Value Transactions (Revenue ≥ $32,400):** Displays transactions meeting the specified revenue condition.
3. **Underperforming Products:** Displays products identified for further performance analysis.

Dashboard 2: Sales Performance Analysis

The dashboard contains the following visualizations:

1. KPI Card – Total Profit.
2. KPI Card – Total Revenue.
3. KPI Card – Profit Margin.
4. KPI Card – Average Revenue per Product.
5. KPI Card – Average Profit per Product.
6. KPI Card – Total Transactions.
7. Clustered Bar Chart – Total Revenue by Product.
8. Pie Chart – Total Revenue by Category.
9. Region Slicer.

Together, these dashboards provide an overview of transaction-level performance and overall sales performance.

4. Results and Observations

- The Top Transactions table helps identify the transactions generating the highest revenue.
- The High-Value Transactions table isolates transactions with revenue of at least $32,400.
- The Underperforming Products table helps identify products requiring further investigation based on revenue and margin.
- Total Revenue and Total Profit KPI cards summarize the financial performance of the business.
- The Profit Margin KPI provides an overview of profitability relative to revenue.
- Average Revenue per Product and Average Profit per Product provide additional measures for comparing product performance.

- The Total Transactions KPI summarizes sales activity.
- The clustered bar chart makes it easier to compare revenue across products.
- The pie chart shows the distribution of revenue across categories.
- The Region slicer supports interactive analysis by allowing users to focus on a selected region.

The dashboards help organize sales data into meaningful tables and visualizations. Actual KPI values and detailed observations can be included based on the results displayed in Power BI.

5. Learning Outcomes

After completing this experiment, I was able to:

1. Import sales datasets from Excel into Microsoft Power BI.
2. Understand and prepare data by checking column names and data types.
3. Apply sorting and filtering techniques to identify high-revenue transactions.
4. Create tables to display high-value transactions and underperforming products.
5. Create KPI cards to display Total Revenue, Total Profit, Profit Margin, Average Revenue per Product, Average Profit per Product, and Total Transactions.
6. Create a clustered bar chart to compare total revenue across different products.
7. Create a pie chart to visualize total revenue distribution by category.
8. Use a Region slicer to filter dashboard data interactively.
9. Analyze sales performance and product profitability using tables, KPI cards, and charts.
10. Develop interactive dashboards to present business insights and support data-driven decision-making.

6. Conclusion

This experiment demonstrates how Microsoft Power BI can be used to import, prepare, sort, filter, analyze, and visualize sales data. The first dashboard focuses on high-revenue transactions, high-value transactions, and underperforming products using tables and filtering techniques. The second dashboard summarizes sales performance through KPI cards, a clustered bar chart, a pie chart, and a region slicer.

The experiment provides practical experience in creating interactive dashboards and presenting business information in a clear visual format. These techniques help users understand sales performance, compare products and categories, examine profitability, and support data-driven business decisions.